

The background of the top half of the cover is a photograph of a forest fire. Several tall, dark tree trunks stand in the foreground, with bright orange and yellow flames and thick white smoke rising behind them. A diagonal green line separates this image from the white text area below.

FOREST REPORT

**A comprehensive Swedish database
for forest and land fires
- actors, information sources and possible future collaboration**

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Preface

Over the past 20 years, the issue of a coherent fire database has been raised several times by various stakeholders and a number of smaller investigations have been carried out, including at the initiative of the Swedish Environmental Protection Agency¹. However, these initiatives have not led to any concrete action plan. In connection with the establishment of the SLU Forest Damage Center in 2021, a proposal was made to review the issue again, and Anders Granström at the Department of Forest Ecology and Management was given the assignment, which was carried out in collaboration with the SLU Environmental Analysis Program Forest and the National Forest Data Lab. In mid-October 2021, a workshop was held in Umeå with representatives from nine organizations with a potential interest in the issue. This report is based, among other things, on contributions during that day, and is intended as a basis for further discussion between the relevant authorities, aiming for a functional, comprehensive database of Swedish vegetation fires.

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¹ Berglund, H. 2015. National IT solution for data on forest fire fields – data sources, flows and costs. PM, Artdatabanken, SLU.

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Summary

Despite the fact that the issue has been raised several times over the past 20 years, there is still no comprehensive database of wildfires and management fires that have occurred in the country, which is unfortunate for both fire protection and nature conservation. This report examines why it is important to have credible statistics in the area, what data collection is done today, and how a national fire database could be created.

The fire database's information on wildfires should primarily be obtained from the incident reports that, according to current regulations, are submitted centrally to the Swedish Civil Contingencies Agency (MSB) after the operation has been completed. However, the incident reports need to be changed so that the starting position of the fire and a shape file for the outer boundary can be included. This information can be added by the rescue service personnel in the field, provided that small modifications are made to the software that the rescue services use. In addition, the MSB requirement specification regarding the incident report needs to be changed.

Information about completed management fires (date of execution and a shapefile for the outer boundary) can be conveniently obtained from the respective land manager. Management burns (nature conservation burns and burns for other purposes) are almost exclusively carried out by

a few large land managers (county administrative boards, forestry companies, the church, the Swedish Fortification Agency, the Swedish Property Agency), so there should be no major problem in agreeing on working routines.

For every wildfire and maintenance fire, there should be information about the prevailing weather and fire risk index is linked. This information can be retrieved via an automatic function at SMHI, provided that the time and position are known. With the help of other already available geographical information (forest condition, soil type, etc.), it will then be possible to quantify what has burned, for what reasons, and what the result was. Not least in the perspective of future climate change, this needs to be monitored over time.

In order for this fire database to be implemented, various stakeholders must agree on a design and a suitable contractor. It is likely that the Swedish Forest Agency will be the contractor, as they have a well-functioning mapping service with relevant background information and are already mapping major fires.

Summary

A national database for wildfires and management fires - actors, sources of information and potential future cooperation

Despite discussions over the last 20 years, there is still no database over wildfires and management fires in Sweden, which hampers both fire protection and nature conservation. This report details the need for accurate statistics in this area, the current status of relevant data sources and potential design of a national wildland fire database.

Information on wildfires could most easily be retrieved from the Incident reports that are sent to MSB (The Swedish civil contingencies agency) after completion of each wildfire incident, and which in their present form contain valuable data.

The structure of these reports would however have to be modified to include also coordinates for the starting point and a shape-file delineating the perimeter of the burnt area, information that the suppression personnel can retrieve before leaving the site.

Information on management fires (date and a shape-file for the perimeter) can most easily be obtained directly from the land managers performing prescribed fires, near-exclusively a few large entities (forest companies, the Swedish Church, Nature reserve managers, the military). A routine for delivering data should be agreed on by all of these.

Current weather and fire danger indices should be linked to each wildfire and management fire.

This can easily be done through an automatic routine at the weather service (SMHI), given that day and position of each event is known. Once these basic pieces of information (time, point of origin, geographical extent, cause, weather and fire danger indices) is available in the database, external geo-information such as fuel type, forest volume etc. can be associated by the user for a range of analyses. A permanent and detailed database will be central for both long-term biomonitoring and wildfire risk assessments, and will be of general interest, not least in the perspective of climate change.

In order for a fire database to be realized, various actors must unite around a concept and determine who is to manage it. The Swedish Forest Agency might be the prime candidate, since they have a well-functioning public map service and since they already undertake a basic mapping of larger burnt forest areas.

1. Background

Many people take it for granted that a country as developed as Sweden today has solid statistics on how much forest land burns per year, and what changes in the number of fires and burned area have occurred over time. This is not the case. After the fires of 2018, for example, the question constantly came up whether these were the worst in 100 years? Yes, probably, but you can't be sure. In fact, the first attempt to collect forest fire statistics was made in the late 1800s.

numbers because fire was then a real threat to forest resources, but the collection only applied to state forests. From the beginning of the 1940s, the fire service also began to collect statistics on its efforts, but interest varied over time and during the period 1980-1996, no registration was actually made at all. Even today, the forest protection aspect is an important reason for collecting statistics, but there is also an interest from nature conservation to follow the extent of wildfires. And in addition, a lot of land is actively burned as a nature conservation measure, which makes it necessary to distinguish in the statistics. So questions about where it burned, for what reasons, which

the effects were etcetera (Figure 1) cannot be answered with the statistical collection that is done today. A comprehensive fire database would therefore be of national interest for both fire protection and nature conservation.

Not least in light of ongoing climate change, it is important to continuously monitor damage to forests, and fires are of course one of the causes of damage. A number of exploratory studies were started in 2021 at SLU's Forest Damage Center, one of which was entitled 'Preliminary study on collaboration around data on burned forests'.

In October 2021, a workshop was held at SLU, Umeå as part of the feasibility study with participation from 9 organizations: SLU, the Swedish Agency for Civil Protection and Emergency Preparedness (MSB), the Swedish Forest Agency, the Swedish Environmental Protection Agency, the SLU Species Data Bank, FSC-Sweden, the Västerbotten County Administrative Board, Region Västerbotten and Holmen Skog AB. During the workshop, the interests of various stakeholders, available data sources and opportunities for collaboration were discussed. This report is based, among other things, on the discussion that took place there.



Figure 1. Examples of questions that can be asked about an individual fire object. The image is from a fire in Östergötland at the end of April 2019. The total area was around 226.5 ha and was determined in this particular case on site with GPS via a careful helicopter flight along the border. The MSB's intervention database (IDA) for this fire states that it affected 2265000 m² (=226.5 ha) 'productive forest land including clearing', while the three other land categories (see Fact Box 1) are set to 0 m². As can be seen in the image, this is not correct, but the fire also spread over a lot of open land, as well as wooded bog and open bog. Using a polygon for the spread of the fire and additional information from different GIS layers, it is possible to automatically answer most of the questions listed in the image. Photo: Anders Granström.

2. How does data collection on wildfires and maintenance fires work today?

2.1. Rescue services and MSB

In accordance with the Swedish Accident Prevention Act (LSO 2003:778), forest and land fires are currently extinguished by municipal rescue services. The third chapter and tenth paragraph of the act states that "When a rescue operation is completed, the municipality shall ensure that the accident is investigated to a reasonable extent to clarify the causes of the accident, the course of the accident and how the operation has been carried out".

This report is made by the operation leader after the operation has been completed in the form of an 'Incident Report', which is sent to MSB's statistics department. MSB is responsible for supporting and developing the municipal emergency services, and maintains a national database for all types of accidents. The database has existed since 1996, but until recently it was a voluntary commitment for municipalities to submit incident reports to it, which most municipalities followed. However, since 1/1/2021 it has been a statutory requirement to submit the reports, and from 1/1/2022 it has also been decided that they must be sent no later than two months after the operation has ended.

A review of the situation carried out at the end of February 2022 shows that the median time for all reports of rescue operations overall was around four days, but that 10% of operations were not reported within 25 days of the event (Figure 2). It is possible that the 'reporting speed' differs somewhat between full-time and part-time stations²

because they later have to do reporting to receive compensation³ It can

² 'Part-time stations' are not permanently staffed. Firefighters have their regular jobs near the station but must report at short notice in the event of an alarm. 'Full-time stations, with permanent staffing and full-time personnel, are only found in larger communities.

³ Information from Christer Björkman, Umeå Fire Department.

For this reason, among other things, it is thought that the lag for reporting forest fires in particular may be shorter than in Figure 2, since a large proportion of the country's forest fires are handled by part-time units. On the other hand, they occur during the summer, with holiday breaks that may have an opposite effect.

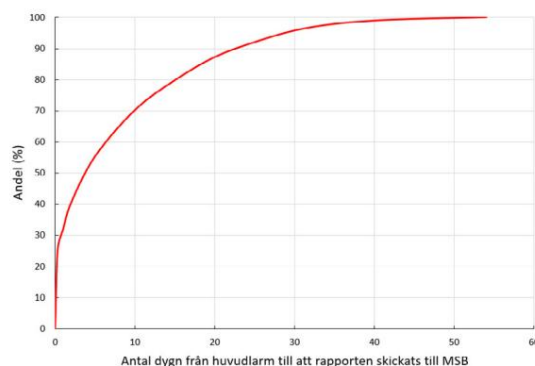


Figure 2. How long does it take for incident reports to be sent? Cumulative distribution of the time from the end of the operation to the incident report being sent to MSB. The material applies to all operations in the country during January 2022 and was selected on February 22 by Joakim Ekberg, MSB. There were probably a number of operations at the time for which the reports had not yet been received, as the maximum time specified is two months. It may therefore happen that the actual distribution for this month's efforts had an even longer tail of late reports.

It would be valuable to review the quality of reporting and potential for improvement for forest and land fires. There is much to be desired, especially regarding the investigation of causes, as around a third of all forest and land fires are noted as "cause unknown" (Sjöström and Granström 2020)!



Figure 3. Each intervention by the rescue service is reported in an incident report, see Fact Box 1. With fairly small modifications to the instructions, the incident report could become much more informative as a basis for a fire database.

Mainly that the fire boundary is laid as a GPS loop, and that a GPS point is set where the fire is believed to have started. In this case, a poorly extinguished campfire on top of a mountain. Via the GPS loop, you automatically get an accurate indication of how large an area has burned, which is otherwise very difficult to assess on site. Photo: Anders Granström.

The incident reports are therefore received by MSB electronically as soon as the report is compiled. MSB does not provide its own software for IT support within the Rescue Service, but each municipality or association of municipalities must purchase it themselves. The market is dominated by the software DAEDALOS from Lunds programarkitekt AB.

The system handles operational efforts as well as finances, personnel, etc. Daedalos is structured in modules and for each specific function there is a 'subscription cost' for the municipality.

Part of the system is DAEDALOS RESPONSE, which is a mobile app with a map function for work in the field, and which can be used by all personnel in the operation (Figure 3). A management support with a corresponding function is CS Atom, from the company CSSuite.

So there is already a skeleton for collecting of geographic information from the field, but minor adjustments must be made to the software to be able to forward the information into the event the report.

The basic information that the incident report should contain has been specified via MSB's regulations (MSBFS 2021.5 Regulations on investigation reports after municipal rescue operations). DAEDALOS has an interface for producing the report, where the incident commander enters a large number of details after the operation has been completed.

Some information is transferred directly from logs made during the SOS operation, while other information must be added manually.

Fact Box 1 lists a number of the data from MSB's regulations that are relevant to forest and land fires, and which therefore end up in the incident report. A critical piece of information is of course the size of the burned area. Here, the area should be stated in square meters (!) for each of four categories: 1) Productive forest land including clearing, 2) Other wooded land, 3) Field or pasture land, 4) Other land without trees. Even correctly distinguishing between some of these four classes can be difficult enough (see Figure 1), but the big problem is still assessing how large an area has burned. Experience shows that rescue service personnel have great difficulty making realistic area assessments in the field. The tendency is for the area to be overestimated, often severely. In a survey of 21 burned areas in Central Norway, it was found that the area for all but four areas had been overestimated and that the median value of the overestimation was a full 3.3 times! (Engström 2000). Although the study was conducted more than 20 years ago, there is plenty of anecdotal evidence that this problem persists. Someone has even coined the term 'rescue hectare'.

However, it is unknown which correction factor should be used today to convert to real hectares.

FACTBOX 1

Examples of information that, according to MSB's regulations, must be included in the incident report after the operation has been completed.

Location (coordinates in Sweref 99)

Times of alarms, decisions about rescue services, arrival at the scene, completion of rescue services, etc.

Information about the origin of the fire

Object that first ignited. First 'object overall', where one choice is 'other'. Under 'other' there are 12 choices, one of which is 'Vegetation'.

Heat source. First, heat source overall, where there are 4 choices, two of which are relevant to fires in vegetation. One of these is 'Fire or flame' and another 'Other'. Under 'Fire or flame' 9 options are listed: # Cigarette, embers from cigarette or similar # Outdoor burning # Gas burner

Gassvets

Glow after burning or grilling

Hygges or nature conservation burning

Match or lighter

Reignition of fire from previous rescue operation # Other fire or flame.

Under 'Other', 9 options are listed, including:

Lightning strike

Friction

Do not spark or ignite

Other heat source.

Cause

Presumed main cause (choice between three main causes, plus 'could not be assessed'). If the main cause was 'human action', there is a follow-up question about assessment of intent.

Original assessment of object that first ignited, heat source or main cause has been updated after investigation (if so click 'yes')

Describe the causes and course of the incident before the arrival of the emergency services (Free text).

After the arrival of the emergency services

Was it burning with flames when the emergency services arrived? (yes, no)

Forest and land fires (specific issues)

Was a fire ban in effect at the time of the fire? (yes, no, don't know)

Burned area upon arrival, or when the rescue service operation has ended, (m2)

Productive forest land including amenity (m2)

Other wooded land (m2). For example, mountain forest, small forest areas in built-up areas, energy forest. # **Field or pasture land** (m2)

Other land without trees (m2). For example, open marshland, heathland or bare mountains.

Parenthetically, it can be mentioned that the worst mis-estimations could be filtered out by a simple comparison with the number of man-hours spent on the operation. When they do not follow a reasonable relationship, it is probably the area that is mis-estimated and not the number of man-hours.

In addition to the information listed in Fact Box 1, the incident reports also contain a lot of information about the total resources deployed in the operation, but hardly anything about how the operation has developed over time or the extent of external resources (landowners, volunteers, flying resources⁴ However, there is room to add detailed descriptions of the process in free text in a few places. Experience has shown, however, that these are often brief, even after really large forest fires (Granström 2020).

In the regulations specifying the investigation reports (MSBFS 2021:5), the fourth paragraph further states the following: "If the municipality assesses that further investigation is needed for

⁴ There is a box to indicate helicopter assistance, but no estimate of scope or impact.
⁵ Small surveillance planes with a crew of pilot and scout. They run predetermined relatively large loops once or sometimes twice a day when there is a high fire risk. It can be assumed that the effective detection distance is 100 miles or more during favorable conditions. When suspicious smoke is detected, SOS is notified and the fire is driven to verify and provide coordinates. The operation is often managed by local flying clubs but is funded by the state via MSB.

to supplement the basic information according to paragraph 3 regarding the causes of the accident, the course of the accident or how the intervention has been carried out, the documentation of each such investigation constitutes a separate part of the investigation report. There are therefore requests in the legislation for an in-depth analysis (so-called §4 investigation) when serious incidents have occurred, but the assessment of whether this has been the case lies with the municipality. Experience shows that this is rarely done for forest and land fires.

2.2 Registration of data at SOS-alarm

Alarms about forest fires come partly from the forest fire brigade⁵ and partly from the public who call 112. SOS creates a case and tries to locate the location as best as possible using all incoming information, often from several callers. The alarm operator follows a pre-established protocol ('Index') to determine the type of fire and the degree of severity (Table 1).

When the position is confirmed, not infrequently at a Overflight by the fire brigade or the first arriving rescue team, coordinates are set for the event. It happens that the same event receives more than one position indication, but then you can filter them out afterwards, although this does not always happen. Some basic information such as time of alarm, arrival at the location of various units, etc. will come later.

Table 1. So-called 'Index' for forest and land fires, i.e. the information that the alarm operator at SOS tries to extract from the caller in order to be able to dimension the alarm from the relevant rescue service, i.e. how much force should be sent. Note that 'football field' is used here, which is also the usual area measurement for forest fires in journalism.

	Type of fire	Characteristics	Alarm level
Fire outdoors	Grass fire	Incandescent fire, no visible flames. Incident, post-inspection	Low
		Detected fire/smoke development	Means
		Fire threatening building	High
	Forest fire	Incandescent fire, no visible flames. Incident, post-inspection	Low
		Detected fire/smoke development	Means
		Area > football field or small fire threatening building	High

to automatically land in the incident report that the rescue leader writes after the operation is completed (see above).

But SOS generates additional information during the operation. They log in real time the position of the rescue service vehicles (with ID), as well as some 'loose' Rakel stations⁶ which is why the commander's position can often be followed, even if they are moving in the terrain. Similarly, the air resources hired by MSB (helicopters and water-bombing aircraft) are logged. This information can be read in the ResQmap map system in the traffic room at SOS, but also in the rescue service's command centers for as long as the operation lasts.

The information about these movements over time during the course of the operation is saved by SOS, which makes it possible to replay the event much later. The rescue service personnel themselves cannot access this information afterwards. However, it is written into the alarm agreement that SOS alarm shall assist the rescue service with this in cases where the municipality wants to carry out an 'enhanced' investigation into an event, i.e. a '\$4 investigation' (see above). Since forest fires are dynamic over time and space and are rarely logged carefully by the rescue service itself, this is a unique source for post-analysis of the event, but is not usually used, not even after major forest fires.

2.3 The Swedish Forest Agency's fire mapping

During the exceptional summer of fires in 2018, it became obvious that a solid area estimate was needed of how much had burned and where. The Swedish Forest Agency then took the initiative to map the fires using satellite scenes (Sentinel-2), or, where applicable, aerial images. The work was carried out during the late summer and autumn of 2018 by several officials who had experience with image interpretation for other purposes, primarily clearing mapping⁷. The goal was to use all of the MSB's incident reports that stated an area of over 0.5 ha, search for the correct position in cloud-free Sentinel scenes taken after the fire, verify that it was a fire and add a shape file for the burned area. A number of the

The largest fires had also been photographed from the air and the mapping was then done from these (Ångra, Enskog, Nötberget, Storbrättan, Fågelsjö, Trängslet).

In recent years (2020-2022), the Swedish Forest Agency has continued along the established path of continuously mapping wildfire areas, and to some extent also management fires. They have then subscribed to alarms about fires in forests and land from SOS. The material has come from SOS in batches once a week and contained information about the alarm time (actually the time when the case was created), position and whether it was a 'forest fire' or 'grass fire'. The classification of fire type is made by the alarm operator and is one of the pieces of information that the alarm operator requests initially (see Table 1).

In addition to the SOS alarms, they have also picked information about active fires from the US fire monitoring system FIRMS (Fire Information for Resource Management System), which uses satellites to monitor the development of fires on a global scale. The system uses data from two different types of satellites: VIIRS and MODIS. The sensors on these register thermal radiation, which makes it possible to detect ongoing fires. There are a couple of overflights per day. The pixel size is 375 m for VIIRS and almost 1 km for MODIS, and a certain minimum radiation energy is needed within the pixel for it to be noted as a fire (via a detection algorithm, where various false signals must also be sorted out). Fires can therefore very well be missed; of course if they have only been active between two satellite passes, but also if they have been too small or low-intensity to be detected.

The information the Swedish Forest Agency retrieves from FIRMS is all 'hot pixels' that have been registered, which for larger fires can be several. To make them easier to find by the fire officer, the map program automatically draws a preliminary figure around the pixels.

From FIRMS come both wildfires and maintenance fires because satellite detection cannot distinguish between these. When it is a verified fire that does not have a nearby SOS alarm, the probability is high that it is a maintenance fire and the object is then registered as such. Potentially, it could occasionally happen that false alarms in connection with maintenance fires give rise to incorrect categorisation of the type of fire. It is not entirely unusual for the emergency services to respond to alarms that turn out to be ongoing natural

⁶ Rakel (Radiocommunication for effective management) is a government network for radio communications. There is usually one Rakel unit per vehicle, but there are also a number of loose, handheld units.

⁷ The Swedish Forest Agency has long carried out an annual check of fellings via 'change detection' in satellite imagery, among other things to detect unreported felling.

It has not been possible to investigate whether there is any screening function to remove such cases from the material sent from SOS.

Another way to misregister is when a conservation burn leads to an actual rescue operation, either in connection with the burn, or a number of days later, as a result of insufficient post-extinguishment. If the fire has then truly spread outside the original area, the burn consists of both a conservation burned area and

partly an area that has been burned by wildfire. In such a case, the entire area will be registered as a wildfire, even if the wild part was only a few square meters, or perhaps remained entirely within the originally burned area, for example in the form of an active smoldering fire that led to an alarm and an emergency call. In such cases, this leads to overreporting of the wildfire area.

The only way to reliably distinguish between wildfires and management fires is to obtain specific information regarding the latter. The Swedish Forest Agency's officers have also tried to obtain information about completed management fires (position and date) through direct contact with forestry companies and county administrative boards, but with varying success.

In the Forestry Board's mapping work, data is loaded. The data from SOS and FIRMS are automatically entered into a specially developed ArcGIS application for verification and processing by handlers (one handler per region; North, Central, South). The handlers then go through alarm after alarm, search Sentinel scenes taken after the given alarm date and draw a polygon around the fire area, if such a thing can be verified (in contrast to pre-scenes, see Figure 4). Alarms that have been marked 'grass fire' have generally been ignored (despite the fact that fires that have been noted by SOS as 'grass fires' in connection with the alarm may well have subsequently spread to forest land).

The information from SOS does not contain any area information and it is also not certain that it was really a fire because false alarms are not filtered out. This means that many alarm points cannot be verified as a real fire. Burned areas must be over a certain size to be reliably seen in a Sentinel scene, especially if it is a dense forest. The pixel size is 10–20 meters, depending on the combination of wavelengths. It should also be borne in mind that the coordinates given by SOS can be at a relatively large distance from the burned area, which makes it difficult to detect small fire fields. The SOS position is often taken from the rescue service vehicle by the road and not out in the terrain at the fire itself.

The Swedish Forest Agency's goal has been to find and map all forest fires >0.5 ha. A large number of real but small fires will therefore not be registered in this mapping.

It should be borne in mind that nearly 90% of all forest

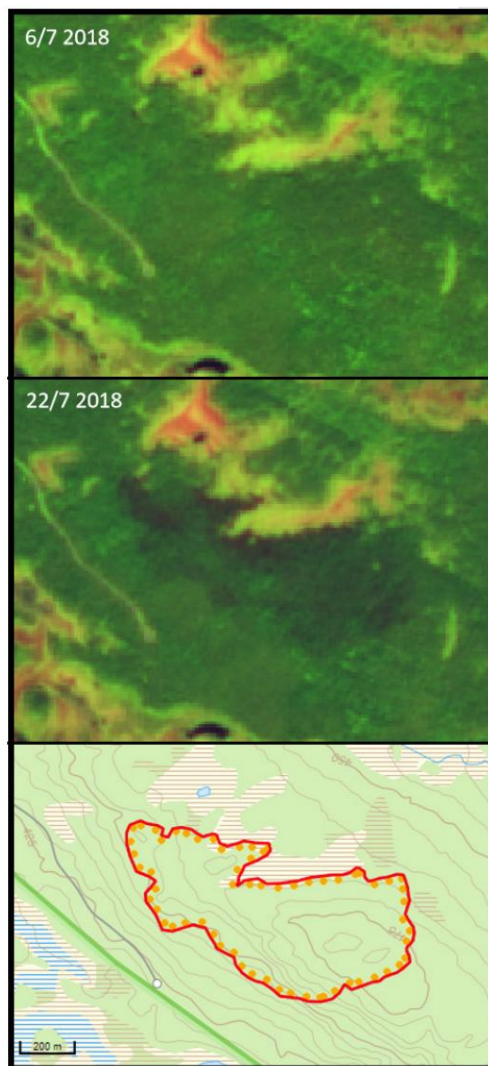


Figure 4. Example of a fire area with drawn fire boundary (forest board's map service), determined via interpretation of satellite scenes (Sentinel-2) taken before and after fire. The area is 37 ha in size and is located NE of Hamra National Park. The fire was started by a lightning strike and was reported on 15/7.

fires have an area⁸ below 0.5 ha (Sjöström and Granström 2020). However, it is unlikely that there is any significant risk of false positive fire registrations, at least not when SOS alarms have been linked to changes in Sentinel scenes.

The subscription to the SOS information cost 30,000 kronor for three months (July–September 8).⁹ The work on the registration has been carried out gradually during the season as time permits and was described by an informant as detective work. The information linked to each fire object in the Swedish Forest Agency's registration is as follows: Fire ID, a polygon, area, date of the fire/burning¹⁰ and a note on whether it is a wildfire or a management fire. There is also space for comments. The Swedish Forest Agency estimates that the total working time for work on the registrations has been around 15 days per year¹¹.

When a fire has been verified, it is directly placed on the Swedish Forest Agency's public map service (subheading Forest damage/Forest fires). The fire limit for all fires registered by the Swedish Forest Agency since 2018 is shown there in red dashes. By selecting different background scenes, such as the latest aerial photo, a user can to some extent see the status of the forest within the area. However, it is not possible to directly find out in the map service which year/date the area burned, the area or whether it was a wild fire or a managed burn. However, the area can be calculated in the map tool itself if you lay out your own polygon (see Figure 4).

2.4 Satellite monitoring of fires within the EU

The EU has built its own satellite-based monitoring system for vegetation fires, EFFIS (European Forest Fire Information System), with approximately the same structure as the US's FIRMS.

The ambition is to register ongoing fires using signals from the two satellite types VIIRS and MODIS, and to map burned areas via

⁸ Based on the MSB's response database, i.e. area estimates by the rescue service's commanders.

⁹ Ahead of the 2022 fire season, the subscription was changed to cover the three months of May-July instead.

¹⁰ For wildfires, the alarm date, which is of course within the fire period, although most of the area may well have burned one or more days later. For maintenance fires detected via satellite, it is only possible to determine the dates between which the burning occurred if you do not see active fire in a Sentinel scene that happens to be taken on the day of the burn.

¹¹ Lennart Svensson, Swedish Forestry Agency, orally.

Sentinel-2, at least for larger fire targets.

During 2020-2022, SMHI and MSB have operated a development project to download VIIRS/MODIS data themselves when the satellites pass, and directly do their own processing.

The aim has been to achieve faster fire detection, by finding fires before they are observed by firefighting aircraft or the public, and thereby being able to alert the rescue services earlier.

The work has not yet been finalised.

The EU's primary goal with fire monitoring is to support firefighting efforts and to quantify the area of burned land from a damage perspective. One of the products is 'burnt area', i.e. an area calculation of individual fire areas, based either on the summed VIIRS pixels or an 'assisted' plot based on Sentinel. It is unclear how much manual work this involves for EFFIS.

It is clear, however, that both of these estimates are extremely approximate and that they may contain many errors. A particularly gross example concerns an area 40 km northeast of Torneträsk where, according to EFFIS, a 447 ha fire is said to have occurred on 19 July 2021¹². The area is a mountain range where no fire impact can be seen in satellite images. The source appears to have been a swarm of incorrect thermal recordings via VIIRS along the mountain slope, which was then built into a 447 ha polygon based on the contrast with the surrounding terrain, seen in Sentinel scenes. This may be an extreme example, but it shows that automatic area interpretations of fire areas need to be verified manually. A simple check could have eliminated this 'fire', since the area assumed to have burned looked the same in Sentinel scenes from the previous year.

EFFIS has no mechanism whatsoever to distinguish between managed fires and wildfires. This means that all burned land

ends up in the same bag and is a priori assumed to be an unwanted fire. Especially for Sweden, where the area of managed burned land on average is possibly on par with the area of wildfires, this leads to distorted statistics, which has been pointed out by the MSB¹³.

2.5 The Swedish Environmental Protection Agency and the county administrative boards

There are a couple of different systems for registering management measures within nature reserves and national parks.

¹² The information remained on the EFFIS map page as late as February 2022.

¹³ Leif Sandahl, verbally.

national parks. Different county administrative boards follow partly different routines. Unfortunately, it is not currently possible to easily obtain basic information (area, polygon, date, fire risk index) for either completed burns or wildfires within formally protected natural areas in the country.

The respective county administrative board naturally has a lot of detailed information about the burns that have been carried out, but this is not necessarily incorporated into any GIS system. In connection with burns, the county administrative board staff usually collect a lot of relevant information (weather and fire risk data before and during the burn, the progression of the burn during the day of the burn, the impact on the stands and the soil). Often a shorter report is compiled and archived locally at the respective county administrative board.

A couple of county administrative boards have, on their own initiative, collected information about planned burning and forest fires in the county. The most extensive database is for Västerbotten and dates back to 1990¹⁴. The ambition has been and still is to include all fires over 1 ha that have occurred within the county. The county administrative board's own burnings are of course included there, and every autumn a question is sent to the forestry companies and other larger landowners to obtain information about burnings and wildfires on their land. The Swedish Forest Agency is also asked, since planned burning on private land is often done with some financial support, which is administered by the Swedish Forest Agency.

2.6 Forestry companies and other large landowners

Larger landowners always have GIS-based registers of their land holdings. The geographical units are 'stands', i.e. geographically delimited areas with relatively uniform status in terms of soil, tree species composition, age and previous treatment. Various management interventions such as felling, soil preparation, etc. are entered in the log for each stand with information about when the action was carried out.

In some companies' systems, there is now room for exact dates, but the resolution has often been months or even years.

The same applies to burning, which is therefore registered for the stand units affected. It is of course possible to see what type of stand was burned and any felling prior to the intervention.

etc., but since the exact date of implementation is usually missing, it is also not possible to add information about weather conditions, drying out, etc., retroactively.

When a wildfire of a sufficiently large area has occurred, the stand map is of course redrawn and new units are created, where some parts of the fire area may be set aside as voluntary reserves while others are felled and then managed as normal stands.

Small fires do not usually result in the stands being redrawn, but a separate 'action area' can be added that follows the boundaries of the burned land instead of the stand boundary. Various potential attributes can be added to the action area such as date, fire risk index, etc.

Although the GIS register itself does not contain If you need more qualified information about the burns that have been carried out, it is easy for the person responsible for forest management to create a polygon for the burned area in GIS immediately after the burn has been carried out, and send the polygon for the burned area to another party. It is then also possible to add an exact fire date and to indicate the purpose of the burn (clearing burning, nature conservation burning).

The easiest way is to just convey the polygon as defined in the stand register, but of course it happens that only parts of the stand that were intended to burn were actually affected by the fire, as well as that the fire at some point went beyond the intended boundary. In the event of major deviations, it can be assumed that a new boundary is drawn, but often the stand boundary must apply as the fire boundary.

At least the larger companies have GIS systems which enables reading cartographic and other information from the database (stock register) in the field on the phone (for example ArcGis Online) as well as directly transferring information from the phone (for example GPS loops) to the stock register (i.e. to the central server), or to another recipient.

When a wildfire occurs on company land, it is As a rule, after completing the extinguishing operation, an employee walks along the fire boundary with a GPS in the field to obtain an exact demarcation of the burn, and this information can then be forwarded with a simple push of a button via email or text message.

Today, companies and larger landowners are not required to report information about management fires to the authorities. Only a few county administrative boards (Västerbotten) request information about management fires from forest companies.

¹⁴ Information from Andreas Garpebring, Västerbotten County Administrative Board.

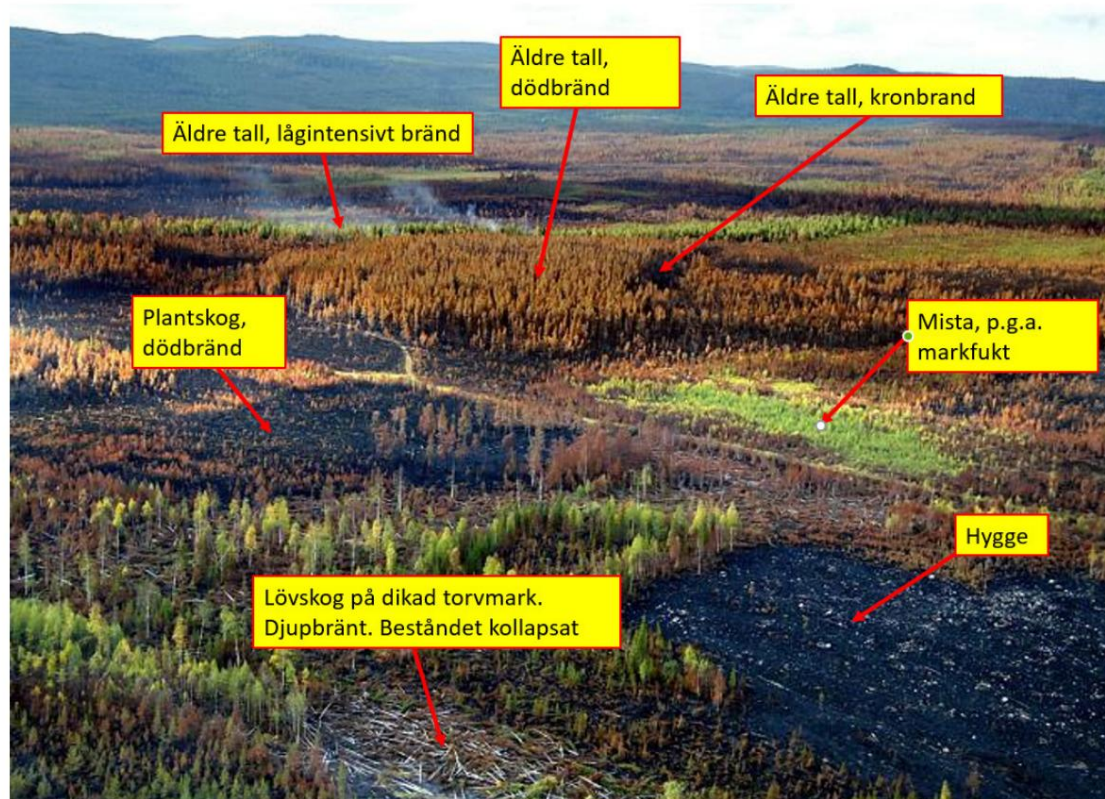


Figure 5. Example of variation in fire behavior and fire effects within the same fire area. If you have a polygon showing the outer limits of the fire, all interpretations suggested in the image can be made more or less automatically via satellite scenes or aerial photos taken after the fire and basic cartographic information from various sources (soil type, tree species distribution, stand age, soil moisture map, fuel map, etc.). Small patches, i.e. remaining unburned patches, can be included in the total fire area for simplicity. The image is from the 1800 ha fire field in Bodträskfors in 2006. Photo: Jon Kanon.

3. Why is a national fire database needed, and who is responsible for it?

asking for information?

Figure 5. Example of variation in fire behavior and fire effects within the same fire area. If one has a polygon showing the outer limits of the fire, all interpretations suggested in the image can be made more or less reasonably. If satellite scenes or aerial images taken after the fire as well as basic cartographic information from various sources (soil type, tree species distribution, stand age, soil moisture map, fuel map, etc.). Small mists, i.e. remaining unburned patches, can for simplicity's sake be included in the total fire area. The picture is from the 1000 ha fire field in Bodträskfors in 2006. Photo: Jon Käreskog.

Several different organizations and agencies need to manage information about forest and land fires. Table 2 briefly lists some of these. The following text provides a more detailed description of the need for information?

and how a fire database can be used.

Table 2. Summary of stakeholders and their needs for a fire database

ACTOR	USE OF FIRE DATA
Rescue services, MSB, county administrative boards	Risk assessment, resource planning, post-analysis, international information exchange
Nature conservation (state and private)	Goal tracking, analysis, planning
Forest certification	Income statement
Landowner	Risk assessment, nature conservation planning
Insurance company	Risk analysis
The Finnish Forestry Board	Damage analysis, consulting
National climate reporting (Swedish Environmental Protection Agency, SLU)	Calculation of carbon flows related to burned land
Research	Risk analyses, biodiversity research, climate-related research

3.1 Rescue services, MSB and county administrative boards to be

Rescue services, MSB as well as the county administrative boards' emergency managers need to be able to easily see the geographical distribution of wildfires using GIS. These must then have a correct starting point and be placed correctly in the terrain, and that the cause, weather and fire risk index are linked to each fire. This makes it possible to both identify and visualize risks in time and space.

In order to be able to perform deeper analyses, it is also necessary to retain the ID from the operation database, which makes it possible to link other more rescue service-specific information from the operation that is stored there, such as resource use, operation times, etc.

From the fire department's perspective, it's not possible to set a specific lower area limit for

which fires should be included in a presumptive fire database. From a prevention point of view and for risk assessment, even small fires are important. The current limit of 0.5 ha used by the Swedish Forest Agency seems unnecessarily high. It means that 88% of all forest fires are screened out (Sjöström and Granström 2020). Ideally, all interventions would be included where it is truly a 'spontaneous' ignition in natural vegetation. If a size criterion is still to be set, 100 m2 may be a more reasonable limit than 5000 m2 (1/2 ha). 100 m2 generally means that there has been at least some spread with blazing fire and that it is not just a fire but a 'proven' potential for spread.

Currently, it is the Swedish Forest Agency's auditor which draws the polygon for fires, which makes it unrealistic to map really small fires, as they are difficult to verify in satellite scenes and

moreover, so numerous. If one can eventually get polygons drawn by the emergency services in the field and incorporated into the incident report, then there is no longer any reason to set a lower size limit.

Incidentally, it is not only forest fires that are of interest to the rescue service, but also those that have affected mainly non-tree-bearing land. Provided that the rescue service can be persuaded to lay a GPS loop around the burned area and forward this with the incident report, the problems with estimating area in the field would disappear, as would the difficulty in distinguishing the four different 'soil types' (see Figure 1 and Fact Box 1). This can instead be done automatically if the polygon is overlaid on relevant map layers that contain soil types.

Of course, there is then the possibility of going much deeper into a statistical analysis of what type of soil or what fuel types are burning, via data such as MSB's fuel map or SLU's tree species map.

A potential use of GIS for the rescue service is the so-called AAR (after action review), that is, reviewing an operation to find shortcomings and success factors. This is rarely done today and one reason is probably that there is no verified and easily accessible geographical information for fires that have occurred. Such reviews are important for really large fires, which by definition are less successful operations (otherwise they would not have become so large). But there is reason to review even successful operations, for example multiple ignitions under high fire risk that were nevertheless able to be kept in place.

The requirements for all such analyses are that there is correct positioning of the ignition point, that the final fire area is known and that there is information about the weather and fire risk indices. In the future, it is also likely that we will have systems for running spatially explicit fire spread models within GIS, which is already being done today in Canada and the USA in connection with major operations (Burman et al. 2016). Such systems may in the future be used by rescue service personnel in Sweden as well, not least to train their understanding of fire behavior on the computer, and to play alternative sequences of events for known major operations (Granström et al. 2023).

MSB is currently funding a three-year development project (2022-2024) to strengthen information

tion management during an ongoing forest fire¹⁵

As the project is now outlined, it will probably be used primarily in connection with large operations that last for a longer period of time. Hopefully, the systems

that are created will make it easier for the operation leader to link basic GIS information (outer boundary, starting point) to the incident report, even after smaller, routine extinguishing operations.

3.2 Nature conservation

County administrative boards as well as the Swedish Environmental Protection Agency have an obvious need to report and illustrate the burnings carried out within formally protected areas, but also to be able to see the wider picture of burnings carried out by other actors as well as wildfires. For certain organisms, the spatial distribution of fires in the landscape is important and there it is important, not least for planning burnings, to be able to see the spatiotemporal pattern of wildfires and completed burnings.

At the national level, there is also a need for reporting to the EU, for example on the conservation status of fire-dependent organisms and specific habitat types, and both the Swedish Environmental Protection Agency and the SLU Species Data Bank have their own interests in following developments over time. Once you have correct polygons for burned areas, it is then relatively easy to follow their further status, for example how much is excluded from forestry in the long term.

It is not only forests that are of interest in this context. The vast majority of 'grass fires' occur on former agricultural land and are of little conservation benefit, but fires in heathlands, mountain terrain and bogs are of interest. For example, at some shooting ranges, burning is carried out annually of largely treeless areas, which in some cases cover hundreds of hectares (Abenius and Larsson 2005; Figure 6).

The aim is to reduce the fuel and thereby reduce the risk of problem fires in connection with shooting. These fires are currently completely outside all statistics, except possibly EFFIS satellite monitoring, if they are detected at all. However, the fact that they are intentional burns is not visible there.

¹⁵ BrandGIS – Management support and geographic situational awareness in the event of forest and land fires. The project is led by Henrik Lundqvist at the County Administrative Board in Kronoberg.



Figure 6. An area at the Mästocka and Tönnersjömålet shooting range in Halland that is burned every spring, and which as a result is home to a unique fauna and flora, including the hairbrush (Abenius and Larsson 2005). Photo: Krister Larsson.

3.3 Forest certification

Both the FSC and the PFSC have certification criteria for larger landowners to burn a certain area of forest land, expressed as a percentage of the harvested area, spread over a five-year period. There are also 'counting rules'¹⁶² in cases where land with a large proportion of remaining standing forest is burned, and land permanently set aside for nature conservation, so-called voluntary set-aside.

There are no regulations requiring landowners to report detailed reports of completed burns to the FSC, only summarized information showing whether or not the goals have been achieved. This also includes areas that have burned in wildfires.

Any review will then be carried out by independent auditors. A fire database would of course facilitate all communication between landowners, certifiers and auditors.

3.4 Landowners

For their conservation planning, the large forest owners actually have the same need as the state conservation agency to be able to see the total spatiotemporal pattern of burned land, i.e. not just what is happening on their own land. Added to this is the need to identify fire risks, as well as to plan

¹⁶² The rules are outlined in the certification criteria. For example, one hectare of burned standing forest is "worth" the same as three hectares of burned clearing.

plan and evaluate their own efforts in connection with wildfires. Even though the rescue services are responsible for the main firefighting, landowners should contribute according to their ability. Not least, it is routine for landowners, especially companies and other larger owners, to carry out post-extinguishment and monitoring of the fire area after the fire has been contained. Even smaller

landowners should be interested in cartographically reviewing fires that have occurred in their immediate area, in order to be able to assess the risk picture and plan their own fire preparedness.

3.5 Insurance companies

The forest insurance market is dominated by a small number of companies and they naturally make premium calculations based on their own claims data. They will presumably also benefit from a general database of fire occurrences, especially if it provides verified area data and information on fire causes. A national fire database also provides a much higher resolution and more accurate picture of the geographical variation of forest fire risk within the country than the data that claims payments can ever provide.

Insurance companies also have an interest in monitoring how the forest fire situation changes over time and how it links to climate, weather-based risk and fire causes.

3.6 Swedish Forestry Agency

In addition to the obvious need to monitor the damage situation in the long term, the Swedish Forest Agency also potentially has other benefits from a fire database. For example, when advising landowners after wildfires have occurred (advice on reforestation, nature conservation provisions, etc.). The Swedish Forest Agency also has an interest in handling grants for nature conservation measures in forestry, often as part of various major EU programs. Here, private landowners have been able to receive financial support for implementing nature conservation burning. The goal monitoring and review of these measures is of course facilitated if there is an easily accessible and up-to-date fire database.

The Swedish Forest Agency also has extensive national responsibility for the production of statistics concerning forests, which are reported online in the 'Swedish Forest Agency's statistics database'. Under the heading 'Measures in forestry, large-scale forestry, recreational and nature conservation burning', data is given per year (since 1997) on the area of land burned. The data is submitted by the respective forest companies as a lump sum for 'Comfort' and 'Nature conservation burning' respectively.

An uncertainty in the statistics is that certain areas land burned by wildfires has been reported to the Swedish Forest Agency as managed fires. It is proven that this occurred in 2014, but has probably occurred in some other years as well. This risks giving rise to double reporting of burned land in climate reporting to the EU and the UN, since this is based on a summation of the MSB's area for wildfires and the Swedish Forest Agency's data on managed fires.

As can be seen in section '2.3 The Swedish Forest Agency's fire mapping', the Swedish Forest Agency has in recent years taken on a national responsibility for mapping both wildfires and management fires and there is a stated ambition to continue, not least in the perspective of climate change and its impact on the forest damage situation in the country.

3.7 National statistical production for climate reporting

For carbon budgeting, among other things, data on burned land is required nationally and internationally.

The responsible authority for climate reporting to the EU and the UN within the "Land Use, Land-Use Change and Forestry" (LULUCF) sector is the Swedish Environmental Protection Agency, but the calculations are made by SLU.

Since the largest instantaneous carbon flows in connection with fire are determined by soil type and degree of drying, it is not sufficient to simply sum up the area of burned land; polygons are needed for each fire area as well as coupled fire risk indices¹⁷ where for example one of them (DMC, duff moisture code) is well correlated with the moisture content in the humus layer and thus to the degree of humus consumption in the fire.

Today, only a general estimate of the carbon flow caused by fire is made, which is based on the area data in MSB's incident reports, inquiries to large-scale forestry about the total area of management burns, and standard assumptions about the impact of fire on the carbon stock.

3.8 Research

A coherent fire database with wildfires and managed burns will facilitate various research initiatives on, for example, fire risk, fire suppression, biodiversity and forest management. In most cases, questions are naturally asked that require further data collection, but selection of sampling areas can often be made based on basic information such as geographical position, date, area, fire risk index, etc.

What information needs to be entered into a database and how can this be done? Table 3 lists a number of basic information that should be added to each wildfire and management fire. Several different ways of accessing the information are conceivable, but the table gives one or a couple of alternatives. Figure 7 shows a possible information flow. Below are some complications for information mainly about wildfires.

¹⁷ In Sweden, the Canadian 'FWI' system is used, which generates an index for calculated relative fire intensity (FWI) and various sub-indices (FFMC, DMC and DC) for moisture content in different fuel layers.

4 A possible future structure

4.1 Wildfires

For each registered wildfire, a fire date is needed. The vast majority of fires are extinguished on the same day they are reported, and should of course be linked in the fire database with the weather and fire risk indices that prevailed that day. However, it is not uncommon for the operation to be formally terminated only the following day, even if there has been no active spread after day one. In cases where the fire has actually spread over more than one day, it is of course necessary, for a qualified analysis, to have data on weather and index values for those days as well. However, it would complicate the database somewhat to include all that information. A compromise is to include the time of the alarm and the time for the rescue service to finish.

In reality, it is even more complicated because the weather and thus the moisture content of the fuel varies within a day. The daily index is always calculated by SMHI at 1 pm solar time and reflects the situation during what is normally the worst fire period during the day. However, since 2021, the fire risk index and weather are also calculated per hour. Potentially, the weather and index for the entire fire period could be linked hour by hour to the respective fire, but this would make the database more complicated.

Furthermore, in reality, there is almost never any high-resolution interpretation of when during the operation the most fire spread occurred, which blocks a deeper analysis. For the sake of simplicity, the daily value can therefore suffice as an indication of weather and drying conditions during each fire.

How then should fire weather and fire risk index can be extracted and linked to individual fires? SMHI continuously calculates this data on a grid size of 2.8 km and the data is saved. Currently, the data is kept available to external users for one year and can be accessed via MSB's fire risk page. Manually extracting data for such a large number of fires as this is the case is unthinkable, but according to SMHI¹⁸ an automatic routine can easily be written that retrieves fire weather and risk indices for individual fires, provided that

ensuring that the fire date and position are known.

This weather and index data is important for various analyses, but there is hardly any rush to obtain it. For example, it could be done at one time at the end of the fire season.

Furthermore, information on the presumed cause of the fire is needed. Here, the information from the response database (Fact Box 1) should be simplified and the cause of the fire should simply be recorded divided into a few but meaningful categories (for example, 'Lightning ignition', 'Machine operation', 'Camp fire', 'Smoking', 'Other known cause', 'Cause unknown'). What the specific causes should be should be further discussed among users. There may be confidential information about certain causes of ignition, for example deliberately set fires (arson) or accidents, which should therefore be placed under 'Other known cause'.

The easiest way to get these basics in Information about wildfires is via incident reporting to MSB and not from SOS alarms. It may take longer to receive the material, but you can get more information and you can avoid the cost of subscribing to SOS alarm.

Two essential pieces of information that are currently missing from the incident report but should eventually be added are the (presumed) starting point of the fire and the final extent of the room (polygon/shapefile). In the future, including this in the incident report requires that the MSB's regulations be reformulated, and that someone in the response team takes a final walk around the fire boundary with GPS, places a mark at the presumed starting point and links the information further. As described under the heading 'Rescue service and MSB', this can be handled with a phone app, after a simple change to the software currently used. In the meantime, the burned area needs to be drawn in manually, as it is today.

As for the starting point, experienced people can often sniff it out with relative certainty, and then it also becomes easier to state a probable cause of the fire in the incident report. Today, 'Cause unknown' is stated.

¹⁸ Johan Böhlén, SMHI, personal communication.

for over 30% of all forest fires. Although every rescue operation already receives a position information in the incident report today, this is often taken from the vehicles on the nearest road (or via the fire aircraft), and is often located a good distance from the actual starting point.

4.2 Maintenance fires

The date of implementation and a polygon for the burned area should be available from the landowner, which in most cases is the county administrative board or one of the large forestry companies, provided that a clear and simple routine for the transfer of information has been agreed upon. Most of the other information can be linked more or less automatically,

Table 3. Basic information that there is reason to collect, or create for wildfires and management fires, and link to a fire database and keep visible in an easily accessible map service, such as the Swedish Forest Agency's. *Table 3. Basic information that there is reason to collect, or create for wildfires and management fires, respectively. An asterisk by incident report means that the information in question is not included as a requirement in the MSB's and linked to a fire database and kept visible in an easily accessible map service, like the Swedish Forest Agency's. 'The incident report*' means that the information in question is not included as a requirement in the MSB's incident reporting regulation, but should be able to be added in the future to strengthen it.*

INFORMATION	POSSIBLE SOURCE, WILD BURNERS	POSSIBLE SOURCE, CARE BURNERS
Date of fire	The incident report	Land managers (companies, reserve managers)
Alarm time, arrival time, end Rtj	The incident report	-
Report number	The incident report (For re-ignitions reported as separate incidents, it is important to link these together)	-
Brandors	Automatic reinterpretation from The incident report	-
Resource usage (man-hours) Event report		-
Polygon surface coverage (shapefile)	Land manager, Swedish Forestry Agency, The incident report*	Land managers (companies, reserve managers), The Finnish Forestry Board
Area	The incident report (for all fires with 'polygon' is calculated total area of the database manager)	Calculated by the database administrator from 'polygon'
Affected land type (forest19, clearing, rock obstacle, bog, open land)	Calculated by the database administrator from 'polygon'	Calculated by the database administrator from 'polygon'
Landowner category	The Finnish Forestry Board	The Finnish Forestry Board
Starting point (Sweref 99)	The incident report*	-
Weather and fire risk index	SMHI via the database manager (Incident report*)	SMHI via the database manager
Brand-ID	The database administrator	The database administrator

¹⁹ Here it is reasonable to make a rough division into, for example, cutting class.

4.2Management fires

The date for implementation and a **polygon** for the burned area should be available from the landowner, which in most cases is the county administrative board or one of the large forestry companies, provided that a clear and simple routine for the transfer of information has been agreed upon. Most of the other information can be linked more or less automatically, such as fire weather and fire risk index (see above under wildfires). It would possibly be of value in the case of company land to provide information on whether the burned area is more or less permanently set aside for nature conservation purposes.

such as fire weather and fire risk index (see above under wildfires). It would possibly be of value in the case of company land to provide information on whether the burned area is more or less permanently set aside for nature conservation purposes.

4.3 General requests for map interface

As described under the heading '2.3 The Forestry Board's fire mapping', the Forestry Board has in its mapping service figures drawn for all fires that they

has mapped. However, it is not possible today to see whether it is a wildfire or not, nor is there a date (not even a year) for the fire/burning.

Regardless of who will manage the fire database in the future, it would be desirable to be able to select 'Year', 'Size category', 'Cause', 'Owner category', with a simple push of a button already in the map service. 'Wild' versus 'Management fires'. In addition, it is desirable that the material can be exported in a simple way.

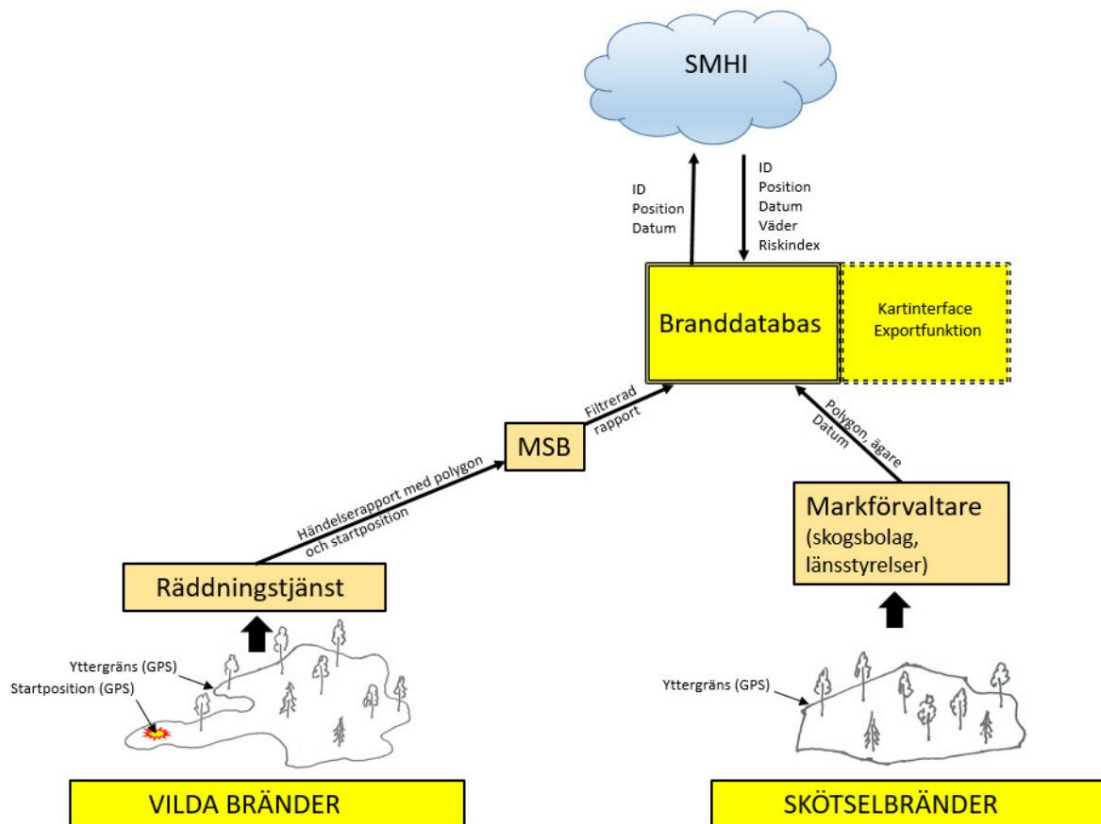


Figure 7. Template sketch for possible information flow to a fire database.

4.4 Conclusions

1. The workshop held in October 2021 showed that there is interest from all participating organizations for a unified fire field database covering both wildfires and maintenance fires. However, it is clear that some steps need to be taken to increase the accuracy of current data collection and make management more rational.

exact coordinates to the client. In order for the information (date, starting point, polygon) to actually reach the fire database, there must be clear instructions on what is being requested, what the purpose is and which account is the recipient. This needs to be anchored with responsible managers within the companies, and other large landowners.
2. The Swedish Forest Agency has good conditions to be both the performer and data host of a fire database.

They have a general responsibility for monitoring forest damage but also for nature conservation work in forests. They also have experienced staff thanks to the mapping they have done in recent years.

If their commitment is to be extended to 'all' fires in forests and land, i.e. without setting any lower area limit and to also include fires in vegetation other than forests, an agreement between the various parties is required and the financing is sorted out.
3. It would be a great improvement if rescue the service could collect spatial data for wildfires directly in the field (starting point and fire boundary) in connection with the termination of the operation and that this data could be automatically incorporated into the incident report. However, this would require both a change in the MSB's regulations and a minor restructuring of the mobile apps used by the rescue services in the field (currently DAEDALOS RESPON and CS Atom).
4. While waiting for a system where the fire service collects and delivers spatial fire data, channels should be developed for the large landowners to send polygons to the Swedish Forest Agency immediately after wildfires have ended. It can be assumed that their personnel almost always GPS-set the fire boundary for their own use if it concerns a significant area of burned land, and not infrequently they can also find a probable starting point if they are asked to do so. However, there is no reason to set a lower size limit. It is true that many small fires will not be coordinated by any official in the field. On the other hand, post-extinguishment and monitoring are almost always carried out even on small fire spots, often by independent contractors, and these patrols can easily forward
5. Today, the Swedish Forest Agency subscribes to relatively 'dumb' basic information (only providing time, position, fire type) from SOS regarding wildfires.

If instead one gains access to the event reports for forest and land fires from MSB, the Swedish Forest Agency's work with mapping will be significantly easier. On the one hand, there is an area data and other information that makes it possible to filter out false alarms etc. and makes it much easier to search for the fire area in satellite scenes. In addition, the event report contains a lot of important information that should be added to the database (ID number, cause, extent of the operation in time, possibly resource use). Since the event reporting system is automatic, it should be possible to filter at the source (MSB) and send an edited version of the reports to the Swedish Forest Agency as soon as they reach MSB. The rules on cooperation between authorities mean that this data should be delivered free of charge, but one must of course carefully evaluate which information should and can be transferred, including taking into account confidentiality rules. Getting basic information from the incident reports instead of from SOS probably results in a certain delay compared to getting SOS data once a week, which has been the case so far in the work of the Swedish Forest Agency. If the curve in Figure 2 is also representative for forest and land fires during the summer, it may still be acceptable. The median time from intervention to reporting was around four days.
6. Management fires can only be captured via direct information from the performers or clients (county administrative boards, forestry companies, dioceses, the Swedish Fortification Agency, etc.). Here, care should be taken to establish communication directly with the district managers at the companies (either the nature conservation specialist or the person responsible for forest management). Today, the districts are so large that relatively few people are involved. The performers of burning

are either our own staff or hired contractors, but the management manager at the administration is always informed when a burn is carried out and has access to the fire date and a polygon for the burn after it has been carried out.

As with wildfires, there is reason to forward the information as soon as possible, which requires a simple and clear routine. There is of course reason for the fire database administrator to check that the given polygon corresponds more or less to what can be read in Sentinel scenes. Should there be clear deviations, for example that a larger part of the area has remained unburned, the given fire boundary should of course be corrected.

7. Weather and fire risk index for the fire day must be linked to each wildfire and management fire, which significantly improves the potential use of the database (Sjöström and Granström 2023). Here it is possible to create a simple Excel spreadsheet that retrieves data via SMHI's API²⁰ based on dates and coordinates. This can either be done continuously or in larger batches. Perhaps the most rational is to do it at a single time, after the end of the fire season in September.
8. The fire database should be made widely available via a map service. The closest is the Swedish Forest Agency's map service, not least because a lot of stand information and other geographical information can be viewed there at the same time. But the fire information must be selectable primarily by year, fire area, and fire type. The exact design should be discussed with potential users from both the rescue and nature conservation sectors.
9. The next step in realizing a national fire database should be a meeting between competent managers at the Swedish Forest Agency, MSB, SMHI, the Swedish Environmental Protection Agency, and the larger forest companies to agree to cooperate on this issue and give the go-ahead for each organization to participate. The authority that takes primary responsibility for the database can then give detailed suggestions on how to structure the information collection.

²⁰ API – application programming interface, the system that enables communication with, for example, a database.

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